ITI CareerTrack

One Permanent Career ID (PCID) that follows every ITI learner from admission to apprenticeship, job, business and higher studies, filled from official records and WhatsApp even when the student does not update it.

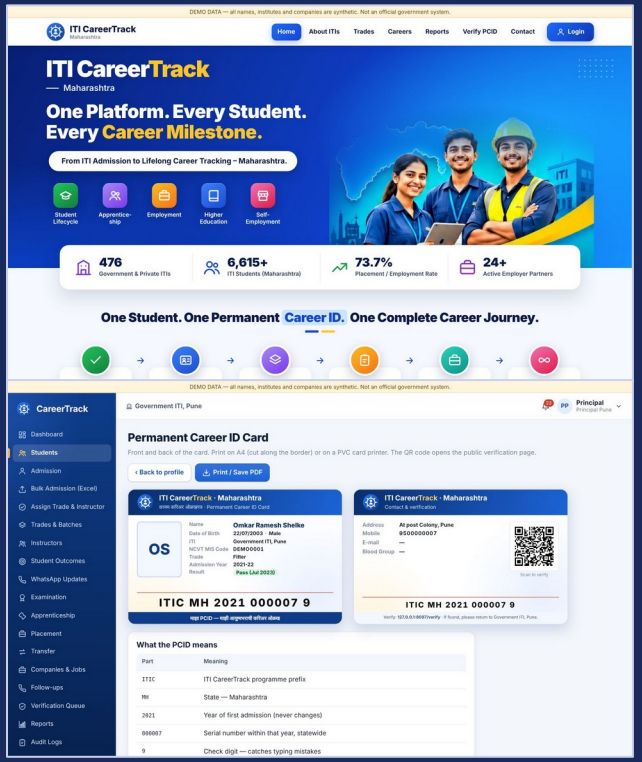
Sub-theme: 5. Apprenticeship, Internship, Placement and Employability Outcome Tracking

Applicant: Sanjay Laxman Rathod (Individual Innovator)

Stage: Working prototype, live with DEMO data

Demo: app.copamcq.com · login DEMO0001 / Demo@1234

For: Approx. 1.52 lakh trainees who appear and 1.37 lakh who pass every year, 430+ Govt ITIs, DVET



Live prototype: home page and PCID card

2

The Problem

What real challenge in society, government service or industry are you solving?

What is the problem?

After the ITI course, nobody can see what a trainee does next. Results, apprenticeships, jobs and studies sit in separate systems, or are not recorded at all. Pass-outs rarely open portals or answer calls.

Who is affected, and where?

ITI trainees and pass-outs across Maharashtra, in 430+ Government ITIs and private ITIs in all 36 districts, many in rural areas. Also principals, District Vocational Training Officers (DVTO), Regional Offices and DVET.

Evidence

Approx. 1.52 lakh trainees appear for the final examination and approx. 1.37 lakh pass every year (DVET / NCVT results, approx.). Data is spread across NCVT MIS, the apprenticeship portal (NAPS), EPFO and colleges, with no common ID.

Current gaps

Manual phone surveys and registers that are slow, incomplete and unverified. Students who cannot be traced are counted as "unemployed". No permanent record follows the learner from one job to the next.

~1.37 lakh

ITI pass-outs every year in Maharashtra whose next step is not tracked (approx. 1.52 lakh appear in exams)

Why it matters

Without outcome data, trades, seats and industry links are planned blind, and the trainees still searching for work get no help.

3 Our Solution & Innovation

What have you built or designed, how does it work, and what is new about it?

Our solution in brief

A working web platform (English/Marathi, mobile). Admission, single or bulk from Excel, issues a PCID and an identity-style PCID card with a QR code. Every exam, apprenticeship, job, business and study is then added to the same PCID, with 8 role-based logins.

How it works

1 Input

Admission data, official lists (NCVT, NAPS, EPFO, Udyam, DTE), WhatsApp replies, employer entries

›

2 Process

Matches every record to the PCID each night (UAN, PCID, Aadhaar hash, mobile + DOB) and updates the status

›

3 Output

Lifelong verified career profile; trade-, ITI- and district-wise outcome dashboards for DVET

What makes it new?

	Existing approach	Our approach
Cost	Manual surveys, registers	Less than ₹5 per learner per year (est.)
Speed / time	Months; once a year	Nightly, automatic
Reach / access	Only those who answer calls	Official records + WhatsApp, rural included
Ease of use	Many portals, Excel	One PCID; reply with a number

Key USP

Tracks learners after they leave the ITI without depending on them: official records are matched to one PCID every night, and rural pass-outs update their status in Marathi on WhatsApp by sending one number. Unknown status is shown as "Not updated", never as unemployed.

4 Feasibility & Implementation

How ready is the solution, and what will it take to implement it on the ground?

Current stage (mark ✔️ where you are today)

Idea — Proof of Concept — ✔️ Working Prototype — Field Pilot — Ready to Deploy

Technology

Software only, no hardware. PHP 8 + MySQL / MariaDB web app, mobile-responsive, WhatsApp Business API. Runs on shared hosting, a VPS or a state data centre. Tested with 459 automated checks and a load test on 2.7 lakh records.

Estimated cost

Prototype already built. Pilot (1 district): about ₹1 lakh (hosting, training, support). State-wide: hosting about ₹2–3 lakh/year plus WhatsApp about ₹55,000/year, i.e. less than ₹5 per learner per year (estimates).

Timeline

3 months: pilot in one district with real ITI data, training principals and instructors.

6 months: whole region; NCVT and NAPS data sharing; WhatsApp live.

12 months: all 430+ Govt ITIs; EPFO / Udyam links; AI-assisted analytics.

Partners

DVET (Skills, Employment, Entrepreneurship & Innovation Dept.) to adopt and pilot; DGT (NCVT MIS, NAPS), EPFO, MSME (Udyam), DTE for data sharing; employers and MSINS.

5

Impact & Scalability

What change will your solution create, and how can it grow beyond the first location?

1.52 lakh

Beneficiaries reached

trainees per year, at state-wide roll-out (approx.)

70%

Pass-outs with verified status

within 12 months (target), compared with no consolidated record today

1.37 lakh

Jobs or livelihoods

pass-outs per year linked to jobs and apprenticeships (approx.)

Social impact

Reaches rural trainees through Marathi WhatsApp, with no app or login needed. Equal visibility for women trainees and every trade. Verified records help learners prove their experience to employers.

Sustainability

Low running cost (under ₹5 per learner per year, estimate), funded from the DVET / department budget. Built to run on a government server. Daily backups and automatic failover.

Scale-up plan

Pilot district → region → all 430+ Govt ITIs and private ITIs. The same model fits polytechnics and other states: only the official data links need to be configured.

6 Team/Individual & Support Required

Who is behind the idea, and what help do you need from the Challenge?

Photo

Sanjay Laxman Rathod

Individual Innovator

Concept, design and development of ITI CareerTrack (end to end)

Already built and tested:

- 8 role logins (State to Student) · Excel bulk admission
- PCID card with QR · automatic linking of official records
- Marathi WhatsApp bot · daily backups and failover
- 459 automated checks passing · load test on 2.7 lakh records

Built as a working prototype with synthetic DEMO data; no official logo or real student data used.

Thank you | sanjwebs@gmail.com | app.copamcq.com

Support we are seeking

- [x] ✔ **Mentorship & technical guidance**
- [x] ✔ **Pilot with a government department**
- [ ] Funding / grant support
- [ ] Incubation & workspace
- [x] ✔ **Testing, certification or IP support**

A DVET pilot in one district with official data access (NCVT, NAPS) would move ITI CareerTrack from prototype to real use for every ITI learner.